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Portfolio construction based on stochastic dominance and target return distributions

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Abstract. Mean-risk models have been widely used in portfolio optimization. However, such models may produce portfolios that are dominated with respect to second order stochastic dominance and therefore not optimal for rational and risk-averse investors. This paper considers the problem of constructing a portfolio which is non-dominated with respect to second order stochastic dominance and whose return distribution has specified desirable properties. The problem is multi-objective and is transformed into a single objective problem by using the reference point method, in which target levels, known as aspiration points, are specified for the objective functions. A model is proposed in which the aspiration points relate to ordered outcomes for the portfolio return. This concept is extended by additionally specifying reservation points, which act pre-emptively in the optimization model. The theoretical properties of the models are studied. The performance of the models on real data drawn from the Hang Seng index is also investigated.

Key words. Portfolio selection – Stochastic dominance – Risk aversion – Multi-criteria optimisation – Reference point method

1. Introduction and Motivation

The portfolio selection problem with one investment period is a particular case of the general problem of choosing between random variables when larger outcomes are preferred. There is, therefore, a need for models that make a choice between random variables (models for preference). Basically, there are three main types of such models: mean-risk models, expected utility maximization and stochastic dominance models.

Mean-risk models are convenient from a computational point of view, but in many cases, depending on the risk measure that is used, they may lack a rational, theoretical basis for making a choice. Thus, the validity of the results provided is often questionable. Expected utility maximisation is usually difficult to put into practice since the choice of a suitable utility function is somewhat subjective. Stochastic dominance has a sounder theoretical basis for making choice under risk. Second order stochastic dominance (SSD) is particularly important in investment problems since it describes the preference of rational and risk-averse decision-makers. The importance of choosing solutions which are non-dominated with respect to stochastic dominance relations is widely recognized.

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Unfortunately, stochastic dominance models are demanding from a computational point of view.

In this paper, we present a model for portfolio selection, which provides a meaningful solution, corresponding to observed economic behavior, and at the same time is practical from a computational point of view. The solution is meaningful in the sense that the selected portfolio is non-dominated with respect to second order stochastic dominance (SSD) and therefore optimal for every rational and risk averse investor. In addition, this portfolio has a return distribution close to a user-specified target (reference) distribution. Thus, this return distribution can be shaped and “crafted” to a desirable form, to the extent that is achievable. In the case of mean-risk models consistent with SSD, the only criterion for selecting a specific portfolio is a desired trade-off between mean return and risk.

In solving the model, there are three possible cases. Firstly, if the reference distribution (appealing as it may seem) is dominated with respect to SSD, the model provides a solution which comes close to this reference distribution, but is SSD efficient (non-dominated). In this case, the achieved return distribution is “better than target”. In the second case, if the reference distribution is non-dominated with respect to SSD, the model provides a portfolio whose return distribution is exactly the reference distribution. Finally, the reference distribution may not be attainable (in the sense that some of its outcomes are too high such that there are no feasible portfolio weights that could produce such a portfolio return distribution). In this case, the model provides a SSD efficient portfolio whose return distribution comes uniformly close to the reference distribution. Our model works also as a test for stochastic dominance: depending on the optimal value of the objective function, it can be established whether the reference distribution is SSD efficient or not.

The rest of the paper is organized as follows. In section two, models for choice under risk are reviewed, together with their advantages and disadvantages. Mean-risk models, expected utility maximization and stochastic dominance models are considered. Section three presents the mathematical formulation of the proposed model. The SSD efficient solutions in the portfolio selection problem are Pareto efficient solutions of a multi-objective linear problem. In order to select a specific solution to this problem (the one that comes closest to the reference distribution), the reference point method (see [10], [24], [26]) is used. This is an interactive method used in multi-objective programming for generating an efficient solution, in which desired levels for the objective functions (called aspiration points) are introduced. The multi-objective problem is transformed into a single objective optimization problem by constructing a scalarizing achievement function, which is then optimized. We present a model where the aspiration points relate to the ordered return outcomes of the portfolio. This model is then extended by additionally introducing reservation points, which act pre-emptively in the optimization. Section four presents the computational results. A data set drawn from the Hang Seng index with price history for 20 stocks over 131 months is used to explore and evaluate the behavior of the models. Section five presents the conclusions.

2. Preliminaries: Models for choice under risk

2.1. The portfolio selection problem

The problem of portfolio selection with one investment period is an example of the general problem of deciding between random variables when larger outcomes are preferred. Decisions are required on the amount (proportion) of capital to be invested in each of a number of available assets such that at the end of the investment period the return is as high as possible. Consider a set of n assets with asset $j \in \{1, \dots, n\}$ giving a return R_j at the end of the investment period. R_j is a random variable, since the future price of the asset is not known. Let x_j be the proportion of capital invested in asset j ($x_j = w_j/w$ where w_j is the capital invested in asset j and w is the total amount of capital to be invested), and let $x = (x_1, \dots, x_n)$ represent the portfolio resulting from this choice. This portfolio's return is the random variable: $R_x = x_1 R_1 + \dots + x_n R_n$ with distribution function $F(r) = P(R_x \leq r)$ that depends on the choice $x = (x_1, \dots, x_n)$.

To represent a portfolio, the weights $(x_1, \dots, x_n)$ must satisfy a set of constraints that forms a feasible set M of decision vectors. The simplest way to define a feasible set is by the requirement that the weights must sum to 1 and short selling is not allowed. For this basic version of the problem, the set of feasible decision vectors is:

$$M = \{(x_1, \dots, x_n) / \sum_{j=1}^n x_j = 1, \quad x_j \geq 0, \quad \forall j \in \{1, \dots, n\}\}$$

Consider a different portfolio defined by the decision vector $y = (y_1, \dots, y_n) \in M$, where y_j is the proportion of capital invested in asset j . The return of this portfolio is given by the random variable $R_y = y_1 R_1 + \dots + y_n R_n$.

The problem of choosing between portfolios $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ becomes the problem of choosing between random variables R_x and R_y . The criteria by which one random variable is considered "better" than another random variable need to be specified and models for choosing between random variables (models for preference) are required. The purpose of such models is firstly, to define a preference relation among random variables and secondly, to identify random variables that are non-dominated with respect to that preference relation.

The next issue is to consider a practical representation for the random variables that describe asset and portfolio returns. We treat these random variables as discrete and described by realizations under T states of the world, generated using scenario generation or finite sampling of historical data. For any $i \in \{1, \dots, T\}$, let state ω_i occur with probability p_i , $\sum_{i=1}^T p_i = 1$. Thus, the random returns are defined on a discrete probability space $\{\Omega, \mathcal{F}, P\}$, with $\Omega = \{\omega_1, \dots, \omega_T\}$, $\mathcal{F}$ a σ -field and $P(\omega_i) = p_i$.

Let r_{ij} be the return of asset j under scenario i , $i \in \{1, \dots, T\}$, $j \in \{1, \dots, n\}$. Thus, the random variable R_j representing the return of asset j is finitely distributed over $\{r_{1j}, \dots, r_{Tj}\}$ with probabilities $p_1, \dots, p_T$. The random variable R_x representing the return of portfolio $x = (x_1, \dots, x_n)$ is finitely distributed over $\{R_{x1}, \dots, R_{xT}\}$, where $R_{xi} = x_1 r_{i1} + \dots + x_n r_{in}$, $\forall i \in \{1, \dots, T\}$.

The main approaches for modelling choice among random variables: mean-risk models, expected utility maximization and stochastic dominance models are now discussed.

2.2. The mean-risk model

Mean-risk models were developed in the early fifties for the portfolio selection problem. In his seminal work "Portfolio selection", Markowitz (see [11]) proposed variance as a risk measure. Since then, many alternative risk measures have been proposed. The question of which risk measure is most appropriate is still the subject of much debate.

In mean-risk models, two scalars are attached to each random variable: the expected value (mean) and the value of a risk measure. Preference is then defined using a trade-off between the mean where a larger value is desirable and risk where a smaller value is desirable:

In the mean-risk approach with the risk measure denoted by ρ , random variable R_x dominates (is preferred to) random variable R_y if and only if:

$E(R_x) \geq E(R_y)$ and $\rho(R_x) \leq \rho(R_y)$ with at least one strict inequality.

Alternatively, we can say that portfolio x dominates portfolio y .

In this approach, the choice x (or the random variable R_x) is efficient (non-dominated) if and only if there is no other choice y such that R_y has higher expected value and less risk than R_x . This means that, for a given level of minimum expected return, R_x has the lowest possible risk, and, for a given level of risk, it has the highest possible expected return. Plotting the efficient portfolios in a mean-risk space gives the *efficient frontier*.

In order to find efficient portfolios, we solve an optimization problem with decision variables $x_1, \dots, x_n$:

$$\begin{aligned} &\text{Min} \quad \rho(R_x) \\ &\text{Subject to:} \quad E(R_x) \geq d \quad \text{and} \quad (x_1, \dots, x_n) \in M \end{aligned}$$

where d represents the desired level of expected return for the portfolio.

Varying d and repeatedly solving the corresponding optimization problems produces the efficient frontier of the mean-risk model.

An alternative formulation, which explicitly trades risk against return in the objective function, is:

$$\begin{aligned} &\text{Max} \quad E(R_x) - \mu\rho(R_x) \quad (\mu \geq 0) \\ &\text{Subject to:} \quad (x_1, \dots, x_n) \in M \end{aligned}$$

Varying the trade-off coefficient μ and repeatedly solving the corresponding optimization problems traces out the efficient frontier.

Mean-risk models are practical from a computational point of view. However, depending on the risk measure to be used, they may lack a rational basis for making choices. For example, consider a mean-risk model with a symmetric risk measure such as the variance and two random variables X and Y . X has one certain outcome W_0 . Y has two outcomes: W_0 and W_1 , each with probability $1/2$. Let $W_1 > W_0$. Obviously Y should be preferred since it yields at least as much as X . However, according to a mean-variance model, neither one is preferred, since, although X has a smaller expected value, it also has zero risk.

2.3. Expected Utility Maximization

The idea of comparing random variables using their expected utilities is quite old. It dates back to at least 1738 (see [1]). However, it was only in the last century and in an economic context, that expected utility theory was extensively used, when apparently reasonable preference axioms were given as a basis for the model. Most significantly, von Neumann and Morgenstern's work (see [13]) introduced the expected utility concept to decision theory.

In expected utility maximization, a single scalar value (the expected utility) is attached to each random variable. Preference is then defined by comparing expected utilities with a larger value preferred.

A *utility function* is a real-valued function defined on real numbers (representing possible outcomes of the random variable), which measures the relative value of outcomes.

Expected Utility Theory provides a basis for extending a utility function defined on real numbers (outcomes) to a utility function defined on random variables. A utility value is assigned to each random variable in terms of the utility values of its outcomes and the probabilities associated with these outcomes:

Given a utility function U , the *expected utility* of a random variable R_x is:

$$E[U(R_x)] = \int_{-\infty}^{\infty} U(\alpha) dF(\alpha),$$

where F is the distribution function of R_x .

In the discrete case, if the random variable R_x has the outcomes $\alpha_1, \dots, \alpha_T$ with probabilities $p_1, \dots, p_T$, the expected utility of R_x is:

$$E[U(R_x)] = p_1 U(\alpha_1) + \dots + p_T U(\alpha_T).$$

The use of utility functions is based on an implied assumption that they reflect the behavior of investors. There are progressively stronger assumptions about the behavior of investors, which lead to increasing requirements for the properties of the corresponding utility functions.

The first assumption on investors' behavior is that they prefer more to less and are hence rational. In order to reflect this non-satiation attitude, the utility function must be non-decreasing. This is the only non-arguable condition for utility functions on wealth since all investors are assumed to be rational.

The second aspect to consider is investors' attitudes towards risk. There are three possible attitudes: risk aversion, risk neutrality and risk seeking. The general assumption is that investors are risk averse, which means that, as wealth increases, each additional increment is less valuable than the previous one. In order to reflect this, the utility function must be non-decreasing and concave (see [4]).

Further assumptions on investors' behavior (e.g. ruin aversion and decreasing absolute risk aversion) are also discussed in the literature (e.g. [4]). These attitudes are less representative of investors' behavior and are not considered further in this paper.

Once a utility function U is chosen, in order to find the efficient portfolios with respect to the expected utility criterion, we solve an optimization problem with decision variables $x_1, \dots, x_n$:

$$\begin{aligned} &\text{Maximise} && E[U(R_x)] \\ &\text{Subject to:} && (x_1, \dots, x_n) \in M \end{aligned}$$

2.4. Stochastic Dominance

The concept of stochastic dominance or stochastic ordering of random variables was inspired by earlier work in the theory of majorization, that is, ordering. It has been used since the early 1950's, in fields such as statistics (see for example [8]). In economics, stochastic dominance was introduced in the 1960's; Quirk and Saposnik (see [19]) considered the first order stochastic dominance relation and demonstrated the connection to utility functions. Second order stochastic dominance was brought to economics by Hadar and Russel (see [5]) and third order stochastic dominance by Whitmore (see [25]); a detailed discussion is given in [7].

A difficulty with expected utility maximization is that the specification of a utility function is a subjective task. Stochastic dominance overcomes this difficulty. With stochastic dominance relations, the random variables are ranked under assumptions about general characteristics of utility functions that follow from prevalent modes of economic behavior. Stochastic dominance ensures that all individuals, whose utility functions are in the same class, rank choices in the same way.

We compare two random variables with respect to stochastic dominance relations by point-wise comparison of some performance functions constructed from their distribution functions. Let F_x and F_y be the cumulative distribution functions of R_x and R_y respectively.

- R_x is preferred to R_y with respect to first order stochastic dominance (FSD) if and only if $F_x(r) \leq F_y(r)$ for every r , with at least one strict inequality.

Notation: $R_x \succ_{FSD} R_y$.

Intuitively, this describes the preference for larger outcomes.

- R_x is preferred to R_y with respect to second order stochastic dominance (SSD) if and only if $F_x^{(2)}(r) \leq F_y^{(2)}(r)$ for every r ,

where: $F_x^{(2)}(r) = \int_{-\infty}^r F_x(t) dt, \forall r \in R$.

Notation: $R_x \succ_{SSD} R_y$.

Alternatively, we can say that portfolio x is preferred to portfolio y with respect to FSD (SSD).

Performance functions for higher order stochastic dominance are similarly defined (see for example [16], [17]).

Stochastic dominance relations are important in decision theory. There are progressively stronger assumptions about investors' behavior that are used in expected utility theory and stochastic dominance, leading to first, second, and higher orders of stochastic dominance relations. It is known that:

- $R_x \succ_{FSD} R_y$ if and only if $E[U(R_x)] \geq E[U(R_y)]$, for every non-decreasing utility function U , with at least one strict inequality. In this case, all rational investors prefer R_x to R_y .

- $R_x \succ_{SSD} R_y$ if and only if $E[U(R_x)] \geq E[U(R_y)]$, for every non-decreasing and concave utility function U , with at least one strict inequality. In this case, all rational and risk-averse investors prefer R_x to R_y (for a proof see [4]).

SSD is able to order more pairs of random variables than FSD. We could have indifference between R_x and R_y with respect to FSD but prefer R_x or R_y with respect to SSD. $R_x \succ_{FSD} R_y$ implies that $R_x \succ_{SSD} R_y$. Thus, an efficient solution with respect to SSD will be also an efficient solution with respect to FSD.

Stochastic dominance relations, and SSD in particular, are the soundest theoretical basis for making rational choices, corresponding to observed economic behavior. Unfortunately, they are difficult to apply in practice. Finding the entire set of efficient solutions with respect to a stochastic dominance relation is intractable in practice when the number of random variables to be compared is not finite (as in the case of the portfolio selection problem).

Consider now the special case where the two random variables R_x and R_y are discrete with T equally probable outcomes. In order to make the comparison, we rank the outcomes of R_x and R_y in ascending order to obtain two T -dimensional vectors: $(\alpha_1, \dots, \alpha_T)$ and $(\beta_1, \dots, \beta_T)$ such that $\alpha_1 \leq \dots \leq \alpha_T$ and $\beta_1 \leq \dots \leq \beta_T$.

$R_x \succ_{FSD} R_y$ if and only if: $\alpha_i \geq \beta_i, \forall i \in \{1, \dots, T\}$, with at least one strict inequality.

$R_x \succ_{SSD} R_y$ if and only if: $\sum_{j=1}^i \alpha_j \geq \sum_{j=1}^i \beta_j, \forall i \in \{1, \dots, T\}$, with at least one strict inequality.

Remark 2.41. In this case, the concept of SSD coincides with the definition of weak majorization (see [12]).

Remark 2.42. SSD is equivalent to generalized Lorenz dominance (see[9], [23]).

Consider a simple example:

$$R_x: P(R_x = 1) = P(R_x = 4) = P(R_x = 3) = P(R_x = 2) = \frac{1}{4}.$$

$$R_y: P(R_y = 3) = P(R_y = 5) = P(R_y = 0) = P(R_y = 2) = \frac{1}{4}.$$

Ordering the outcomes of R_x and R_y we obtain two 4-dimensional vectors: $(1, 2, 3, 4)$ and $(0, 2, 3, 5)$ respectively. Neither R_x nor R_y dominates with respect to FSD. If we cumulate the outcomes of these vectors, we obtain the vectors: $(1, 3, 6, 10)$ and $(0, 2, 5, 10)$, respectively. Since all the components of the first vector are greater than or equal to those of the second one, it follows that R_x dominates R_y with respect to SSD. This was to be expected, since R_x has a higher outcome than R_y at the lowest level at the expense of a lower outcome at the highest level. Thus, a risk averse investor will prefer R_x to R_y .

3. A model for portfolio selection

3.1. SSD efficiency and multi-objective optimization

Consider a portfolio $x = (x_1, \dots, x_n)$ with return R_x and the case of T scenarios occurring with probabilities $p_1, \dots, p_T$. The realization of R_x under scenario $i \in \{1, \dots, T\}$ is: $x_1 r_{i1} + \dots + x_n r_{in}$, where r_{ij} is the return of asset j under scenario i . Consider further, the case of equally probable scenarios:

$$p_i = \frac{1}{T}, \quad \forall i \in \{1, \dots, T\}.$$

The assumption of equiprobable scenarios is not restrictive in the practice of portfolio selection. Moreover, non-equal probabilities can be handled, for example, by multiple replications of some scenarios - in the case where probabilities are given as rational numbers.

Since we search for solutions x such that R_x has outcomes as large as possible under each scenario, the portfolio selection problem may be viewed as a multi-objective problem with T objective functions to maximize:

$$\max\{f(x) = (f_1(x), \dots, f_T(x)) : x \in M\} \quad (1)$$

where $f_i(x) = x_1 r_{i1} + \dots + x_n r_{in}$ is the portfolio return under scenario $i \in \{1, \dots, T\}$ and M is the set of feasible decision vectors.

$y \in \mathbb{R}^T$ is called an *achievement vector* if there exists a feasible solution $x \in M$ such that $y = f(x)$.

Usually, in multi-objective optimization, two achievement vectors are compared using the Pareto preference relation, defined as follows:

An achievement vector $y = (y_1, \dots, y_T)$ *Pareto dominates* another achievement vector $y' = (y'_1, \dots, y'_T)$ if $y_i \geq y'_i, \forall i \in \{1, \dots, T\}$, with at least one strict inequality.

x_0 is a *Pareto efficient (non-dominated) solution* of (1) if and only if there does not exist a feasible x such that $f(x)$ Pareto dominates $f(x_0)$. In other words, a Pareto efficient solution is a feasible solution such that, in order to improve upon one objective function, at least one other objective function must assume a worse value.

However, in the portfolio optimization problem, in addition to Pareto dominance, there are other considerations. Thus, we do not look for simply Pareto efficient solutions.

Firstly, an achievement vector represents a random variable with T equally probable outcomes for which the ordering is immaterial. We do not compare vectors, but the distributions that these vectors represent. We compare distributions with respect to the first order stochastic dominance relation (FSD) by comparing the k -th worst outcomes of the achievement vectors, for $k = 1, \dots, T$. Thus, the distribution represented by y dominates that of y' with respect to FSD if the k -th worst outcome of y is greater than the k -th worst outcome of y' , $\forall k \in \{1, \dots, T\}$ (see section 2.4).

Secondly, if we compare distributions from the point of view of a risk-averse decision-maker, we consider a surplus more valuable at low levels of outcomes than at high levels. Thus, transferring a small amount from an outcome to any worse outcome results in a preferred achievement vector, since it represents a less risky distribution. To reflect this, we compare two achievement vectors by comparing their cumulative outcomes (i.e., the sums of their worst k outcomes, for each $k \in \{1, \dots, T\}$) (see [14], [15]). In terms of distributions, this means comparing the two distributions represented by those vectors with respect to second order stochastic dominance. Thus, the distribution represented by y dominates that of y' with respect to SSD if the sum of the worst k outcomes of y is greater than the sum of worst k outcomes of y' , $\forall k \in \{1, \dots, T\}$ (see section 2.4).

Remark 3.11. The Value-at-Risk (VaR) at confidence level $\frac{k}{T} \in (0, 1)$ of a random variable R_x is defined as: $VaR_{\frac{k}{T}}(R_x) = -\inf\{z : P(R_x \leq z) > \frac{k}{T}\}$. It easily follows that for a random variable with T equally probable outcomes, the k -th worst outcome is

minus VaR at confidence level $\frac{k}{T}$. Thus, in this case, a random variable dominates another one with respect to FSD if its VaR at confidence level $\frac{k}{T}$ is smaller, $\forall k \in \{1, \dots, T\}$.

Remark 3.12. For a random variable with T equally probable outcomes, the Conditional Value-at-Risk (CVaR) at confidence level $\frac{k}{T} \in (0, 1)$ is minus the mean of its worst k outcomes (this follows from the CVaR definition of Rockafellar and Uryasev [20], [21]). Thus, in this case, a random variable dominates another one with respect to SSD if its CVaR at confidence level $\frac{k}{T}$ is smaller, $\forall k \in \{1, \dots, T\}$.

In order to find the FSD efficient solutions of the portfolio selection problem, the objective functions in (1) are replaced by the k -th worst portfolio returns, for $k = 1, \dots, T$ respectively (see [15]).

Of greater interest are the SSD efficient solutions, which are obtained by replacing the objective functions in (1) with the cumulative outcomes: sums of the worst k portfolio returns, for $k = 1, \dots, T$.

The following result expresses the cumulative outcomes of a vector as the optimal value of a LP problem:

For every $k \in \{1, \dots, T\}$, the sum of the worst k outcomes of a given achievement vector $y = (y_1, \dots, y_T)$ is the optimal value of the following LP problem:

$$\begin{aligned} \max \quad & (kt_k - \sum_{i=1}^T d_{ki}) \\ \text{Subject to:} \quad & t_k - y_i \leq d_{ki}, \quad \text{for } i = 1, \dots, T \\ & d_{ki} \geq 0, \quad \text{for } i = 1, \dots, T \end{aligned}$$

The proof follows easily from the CVaR optimization formula of Rockafellar and Uryasev (see [20], [21]). An independent proof is given in [15].

For each $k \in \{1, \dots, T\}$, the above problem has one free variable t_k and T non-negative variables d_{ki} , $i = 1, \dots, T$. The value of t_k is the k -th worst outcome of the achievement vector y (see [15], [20]). For each $i \in \{1, \dots, T\}$, d_{ki} takes value 0 if y_i is greater than or equal to the k -th worst outcome t_k ; otherwise, $d_{ki} = t_k - y_i$. This means that at most $(k - 1)$ values of d_{ki} are strictly positive.

Thus, we can express the SSD efficient solutions for the portfolio selection problem as Pareto efficient solutions of the following multi-objective linear problem:

$$\text{Max} \quad (t_1 - \sum_{i=1}^T d_{1i}, 2t_2 - \sum_{i=1}^T d_{2i}, \dots, Tt_T - \sum_{i=1}^T d_{Ti}) \quad (2)$$

Subject to:

$$\begin{aligned} t_k - f_i(x) &\leq d_{ki}, \quad \text{for } i, k = 1, \dots, T \\ d_{ki} &\geq 0, \quad \text{for } i, k = 1, \dots, T \\ \sum_{j=1}^n x_j &= 1 \\ x_j &\geq 0, \quad \text{for } j = 1, \dots, n \end{aligned}$$

where $f_i(x) = \sum_{j=1}^n r_{ij}x_j$ is the portfolio return in scenario i , $i = 1, \dots, T$.

Proposition 3.11. A portfolio $x^* = (x_1^*, \dots, x_n^*) \in M$ is SSD efficient if and only if there exist $t^* = (t_1^*, \dots, t_T^*)$ and $d^* = (d_{ki}^*)_{k,i=1,\dots,T}$ with $d_{ki}^* \geq 0$, such that (x^*, t^*, d^*) is a Pareto efficient solution of (2).

Proof. Let $x^* = (x_1^*, \dots, x_n^*) \in M$ be a SSD efficient portfolio. This is equivalent to saying that $(f_1(x^*), \dots, f_T(x^*))$ is SSD efficient among the set of random variables expressing portfolio returns. For every k in $\{1, \dots, T\}$ we solve the optimization problem with variables t_k and d_{ki} , $i = 1, \dots, T$.

$$\begin{aligned} &\text{Max} && (kt_k - \sum_{i=1}^T d_{ki}) \\ &\text{Subject to:} && t_k - f_i(x^*) \leq d_{ki}, \quad \text{for } i = 1, \dots, T \\ &&& d_{ki} \geq 0, \quad \text{for } i = 1, \dots, T \end{aligned}$$

Let t_k^* and d_{ki}^* denote the optimal solution. $kt_k^* - \sum_{i=1}^T d_{ki}^*$ is the sum of the worst k outcomes of $(f_1(x^*), \dots, f_T(x^*))$.

For $t^* = (t_1^*, \dots, t_T^*)$ and $d^* = (d_{ki}^*)_{k,i=1,\dots,T}$, we claim that (x^*, t^*, d^*) is a Pareto efficient solution of (2). Indeed, assume that there exists a feasible solution (x', t', d') of

(2) that Pareto dominates (x^*, t^*, d^*) . This means that: $kt_k' - \sum_{i=1}^T d_{ki}' \geq kt_k^* - \sum_{i=1}^T d_{ki}^*$, for every $k \in \{1, \dots, T\}$, with at least one strict inequality. This, in turn, implies that, the sum of the worst k outcomes of $(f_1(x'), \dots, f_T(x'))$ is greater than or equal to the sum of the worst k outcomes of $(f_1(x^*), \dots, f_T(x^*))$, with at least one strict inequality. This contradicts the fact $(f_1(x^*), \dots, f_T(x^*))$ is SSD efficient.

For the converse, let (x^*, t^*, d^*) be a Pareto efficient solution of (2). The proof that x^* is an efficient portfolio is equivalent to $(f_1(x^*), \dots, f_T(x^*))$ being SSD efficient among the set of random variables expressing portfolio returns. Suppose that there exists another portfolio x' such that $(f_1(x'), \dots, f_T(x'))$ dominates $(f_1(x^*), \dots, f_T(x^*))$ with respect to SSD. This means that the sum of the worst k outcomes of $(f_1(x'), \dots, f_T(x'))$ is greater than or equal to the sum of the worst k outcomes of $(f_1(x^*), \dots, f_T(x^*))$, with at least one strict inequality.

For every $k \in \{1, \dots, T\}$ we solve the optimization problem with variables t_k and d_{ki} , $i = 1, \dots, T$.

$$\begin{aligned} &\text{Max} && (kt_k - \sum_{i=1}^T d_{ki}) \\ &\text{Subject to:} && t_k - f_i(x') \leq d_{ki}, \quad \text{for } i = 1, \dots, T \\ &&& d_{ki} \geq 0, \quad \text{for } i = 1, \dots, T \end{aligned}$$

Let by t_k' and d_{ki}' denote the optimal solution. For every k in $\{1, \dots, T\}$, $kt_k' - \sum_{i=1}^T d_{ki}'$ expresses the sum of the worst k outcomes of $(f_1(x'), \dots, f_T(x'))$. It follows that

$kt'_k - \sum_{i=1}^T d'_{ki} \geq kt_k^* - \sum_{i=1}^T d_{ki}^*, \forall k \in \{1, \dots, T\}$ with at least one strict inequality, which contradicts the fact that (x^*, t^*, d^*) is a Pareto efficient solution of (2).

The problem (2) has T free variables $t_1, \dots, t_T$, with t_k being the k -th worst outcome of the portfolio return distribution, T^2 non-negative variables $(d_{ki})_{k,i=1,\dots,T}$ with the interpretations given above and n non-negative variables giving the optimal portfolio weights.

Remark 3.13. The k -th objective function in (2) represents the sum of the worst k outcomes of $(f_1(x), \dots, f_T(x))$. The mean of the worst k outcomes is obtained by dividing the k -th objective function in (2) by k : $t_k - \frac{1}{k} \sum_{i=1}^T d_{ki}$. Replacing the objective functions in (2) with $(t_1 - \frac{1}{1} \sum_{i=1}^T d_{1i}, t_2 - \frac{1}{2} \sum_{i=1}^T d_{2i}, \dots, t_T - \frac{1}{T} \sum_{i=1}^T d_{Ti})$, we obtain a multi-objective model equivalent with (2), in the sense that both models have the same set of efficient solutions.

3.2. The reference point method

The problem (2) has an infinite number of efficient solutions and additional criteria are needed in order to select a specific solution.

By specifying desirable outcomes for the portfolio return under the various scenarios, we are specifying a desirable (target) distribution. This results in desirable outcomes for the cumulative returns, and hence desirable outcomes for the objective functions in (2). We want to find the asset weights $x_1, \dots, x_n$ that produce a SSD efficient distribution which comes close to the target distribution.

The method used is the Reference Point Method ([10], [26]). In order to select a Pareto efficient solution of a multi-objective problem, aspiration levels (called reference points) are specified for some or all of the objective functions in the multi-objective problem. The aspiration level for an objective function must lie between the lower and the upper bound of that objective function. Depending on these reference points, a special scalarizing achievement function is constructed, which, when maximized, produces an efficient solution of the original multi-objective problem. Thus, the multi-objective problem is transformed into a single-objective problem.

Moreover, the Pareto efficient solution obtained results in objective function values that are:

- Uniformly better than the reference point, if the reference point is not a Pareto efficient achievement vector of the multi-objective problem; they are obtained from the reference point by increasing some of the outcomes until Pareto efficiency is obtained. In this case, the maximization of the achievement function has a positive optimal value.
- Exactly the reference point, if the reference point is a Pareto efficient achievement vector of the multi-objective problem. In this case, the maximization of the achievement function has a zero optimal value.

- Uniformly close to the reference point, if the reference is not attainable. In this case, the maximization of the achievement function has a negative optimal value.

The process is interactive in that a solution is presented to the decision maker who can modify the aspiration levels in the light of the solutions already presented and obtain a further candidate solution.

The Reference Point Method has been seen as a generalization of goal programming. It preserves the main advantage of goal programming: the appealing idea that we set a goal in the objective space and try to come close to it. However, it does more than goal programming, because the solution efficiency is guaranteed: when the goal (reference point) itself is not a Pareto efficient achievement vector, the method improves on the reference point. As opposed to goal programming, optimization continues even after the goal has been reached. The meaning of “coming close” is thus not the traditional one, but “coming close or better” ([10]).

Let $z^a = (z_1^a, \dots, z_T^a)$ be the target (i.e., z_k^a is the aspiration level for the k -th objective function). The simplest form of achievement function is:

$$\gamma_{z^a}(z) = \min_{k \in \{1, \dots, T\}} (z_k - z_k^a) + \varepsilon \sum_{k=1}^T (z_k - z_k^a) \quad (3)$$

where $z = (z_1, \dots, z_T)$ is the current achievement vector and ε is an arbitrary small parameter.

The terms $(z_k - z_k^a)$ in (3) are usually replaced with more complicated functions of z_k and z_k^a , $\gamma_k(z_k, z_k^a)$, which must satisfy certain properties: being monotonically increasing with respect to z_k and taking value 0 if $z_k = z_k^a$ (see for example [10]). These functions are called *partial achievement functions* since they measure the actual achievement of outcome z_k with respect to its corresponding aspiration level z_k^a .

Various functions $\gamma_k(z_k, z_k^a)$ provide a wide modelling environment for measuring individual achievements. Another example of such functions $\gamma_k(z_k, z_k^a)$ are piece-wise linear functions with two segments whose gradients depend on whether the current outcome is above or below its aspiration point (see [10]).

Remark 3.21. The scalarizing achievement function is essentially defined by the worst partial achievement: $\min_{k \in \{1, \dots, T\}} \gamma_k(z_k, z_k^a)$, which in our case means

$\min_{k \in \{1, \dots, T\}} (z_k - z_k^a)$. The latter term is introduced in order to guarantee the solution efficiency in the case where the maximization of the worst partial achievement has non-unique optimal solutions.

Remark 3.22. Different values of ε can lead to different efficient solutions. A solution that maximizes γ_{z^a} will be a Pareto efficient solution of the multi-objective problem with a bound of $M = 1 + \frac{1}{\varepsilon}$ on trade-off coefficients (see [10] for more details).

In addition, one could introduce reservation levels z_k^r for the k -th objective function. In this case, the partial achievement functions are $\gamma_k(z_k, z_k^a, z_k^r)$.

The difference between aspiration and reservation levels is, as the name implies, that whilst it is desirable to achieve aspiration levels, reservation levels should be achieved if

at all possible. The scalarizing achievement function must ensure that reservation levels pre-empt aspiration levels. Thus, reservation levels represent soft lower bounds of the objective functions.

In this paper, we consider piecewise linear partial achievement functions that use both aspiration and reservation levels (see [6], [10]):

$$\gamma_k(z_k, z_k^a, z_k^r) = \begin{cases} \frac{\alpha(z_k - z_k^r)}{z_k^a - z_k^r}, & \text{for } z_k \leq z_k^r; \\ \frac{z_k - z_k^r}{z_k^a - z_k^r}, & \text{for } z_k^r < z_k < z_k^a; \\ \frac{\beta(z_k - z_k^r)}{z_k^a - z_k^r} + 1, & \text{for } z_k \geq z_k^a \end{cases} \quad (4)$$

where α, β are arbitrarily defined parameters such that $0 < \beta < 1 < \alpha$.

The parameter $\alpha > 1$ represents the decision maker's dissatisfaction for outcomes worse than their corresponding reservation level, while the parameter $\beta < 1$ represents satisfaction for outcomes better than their aspiration level ([6]).

The partial achievement functions (4) have attractive properties, which allow a simple interpretation of results. They take:

- negative values if the outcome is below its reservation level ($z_k < z_k^r$);
- value 0 for $z_k = z_k^r$;
- value between 0 and 1 for outcomes between reservation and aspiration level;
- value 1 for $z_k = z_k^a$;
- value greater than 1 for outcomes better than the aspiration level.

Thus, they map outcome values onto a normalized scale of decision maker's satisfaction ([6]).

Moreover, under the condition $0 < \beta < 1 < \alpha$, the partial achievement function (4) is strictly increasing and concave (see Figure 1); thus, it is a typical utility function for a rational and risk-averse decision-maker, and essentially the problem is one of expected utility maximization.

Hence, it can be expressed in the form:

$$\gamma_k(z_k, z_k^a, z_k^r) = \min \left(\frac{\alpha(z_k - z_k^r)}{z_k^a - z_k^r}, \frac{z_k - z_k^r}{z_k^a - z_k^r}, \frac{\beta(z_k - z_k^r)}{z_k^a - z_k^r} + 1 \right) \quad (5)$$

which is useful when writing the model, since we can express the partial achievement functions using linear constraints:

$$\begin{aligned} \gamma_k(z_k, z_k^a, z_k^r) &\leq \frac{\alpha(z_k - z_k^r)}{z_k^a - z_k^r}; \\ \gamma_k(z_k, z_k^a, z_k^r) &\leq \frac{z_k - z_k^r}{z_k^a - z_k^r}; \\ \gamma_k(z_k, z_k^a, z_k^r) &\leq \frac{\beta(z_k - z_k^r)}{z_k^a - z_k^r} + 1. \end{aligned}$$

Since the achievement function is maximized, the largest possible values of $\gamma_k(z_k, z_k^a, z_k^r)$ are required. Thus, the linear constraints above are equivalent to (5).

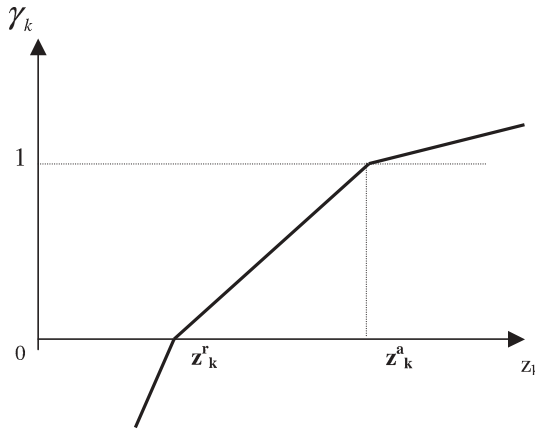


Fig. 1. Example of a piecewise linear partial achievement function (4)

Remark 3.23. When the aspiration and the reservation levels for an objective function are chosen, they must lie between the lower and the upper bound of that objective function. So, if $z_k^{\min}$ and $z_k^{\max}$ are the worst and best values for the k -th objective function, respectively, we require:

$$z_k^{\min} \leq z_k^r \leq z_k^a \leq z_k^{\max} \quad (6)$$

Provided condition (6) is satisfied for every $k \in \{1, \dots, T\}$, the maximization of the scalarizing achievement function produces an efficient solution of the initial multi-objective problem (2) (see [10]).

Furthermore, the Pareto efficient solution obtained results in outcomes (objective functions) that are:

- Uniformly better than the aspiration point, if the aspiration point (and thus the reservation point) is not a Pareto efficient achievement vector of the multi-objective problem. In this case, the maximization of the achievement function has an optimal value strictly greater than one.
- Exactly the aspiration point, if the aspiration point is a Pareto efficient achievement vector of the multi-objective problem (thus the reservation point is not Pareto efficient). In this case, the maximization of the achievement function has an optimal value equal to one.
- Uniformly better than the reservation point and uniformly close to the aspiration point, if the reservation point is not Pareto efficient and the aspiration point is not attainable. In this case, the maximization of the achievement function has an optimal value between zero and one.
- Exactly the reservation point, if the reservation point is a Pareto efficient achievement vector of the multi-objective problem (thus the aspiration point is not attainable). In this case, the maximization of the achievement function has a zero optimal value.
- Uniformly close to the reservation point, if the reservation point (and thus the aspiration point) is not attainable. In this case, the maximization of the achievement function has a negative optimal value.

The converse is true, in the sense that for every efficient solution, there exist aspiration (and reservation) levels such that this efficient solution maximizes the corresponding achievement function (see [10]).

3.3. Models for portfolio selection

Over the last thirty years, there have been attempts to combine the practicality of mean-risk models with the theoretically solid basis of stochastic dominance. Most of these attempts resulted in mean-risk models that provide solutions which are non-dominated with respect to stochastic dominance relations (see for example [3], [16], [17], [22], [27], [28]). Some open questions need to be considered. Firstly, which risk measure—among those consistent with stochastic dominance—should be used? The question of which risk measure is most appropriate is still the subject of much debate. Secondly, and more importantly, from the set of efficient solutions of a specific mean-risk model, which solution should be actually implemented? Not every solution efficient with respect to stochastic dominance is satisfactory. As an example, the portfolio consisting of just the asset with the highest expected value of return is SSD efficient; it is widely recognized that this is not a good choice, due to lack of diversification. Additional criteria for choice are needed. With mean-risk models, one can make a choice simply based on a desired level of trade-off between mean return and risk. This criterion offers limited opportunity to model the distribution of the solution.

In this section, two models for portfolio selection (models M1 and M2) are presented. The motivation for model M1 is to find a SSD efficient portfolio whose return distribution comes close to a “target” aspiration distribution. Model M2 considers also a “reservation” distribution, which represents a first level target.

For model M1, let asp_k be the aspiration level for the k -th objective function in (2), which is the k -th cumulative outcome (sum of the worst k outcomes) of the return distribution that results, $k = 1, \dots, T$. These levels imply aspiration levels for the ordered outcomes themselves and hence define a distribution, which we refer to as the “target”, “reference” or “aspiration” distribution.

Intuitively, the stochastically larger random variable has a smaller distribution function (see Figure 2). Thus, our aim is to obtain a distribution whose outcomes are positioned further to the right.

We keep the same notation used in (2). In addition, let z_k represent the sum of the worst k outcomes, for $k = 1, \dots, T$, and let $\delta = \min_{k=1, \dots, T} (z_k - asp_k)$ represent the worst partial achievement. z_k and δ are free variables.

The following model for portfolio optimization uses the achievement function defined by (3).

$$\begin{aligned} \max \quad & \delta + \varepsilon \left(\sum_{k=1}^T z_k - \sum_{k=1}^T asp_k \right) \\ \text{Subject to:} \quad & \end{aligned} \tag{M1}$$

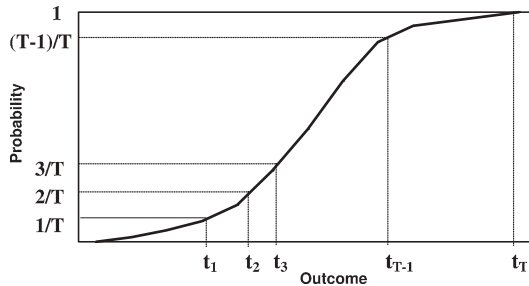


Fig. 2. Cumulative distribution function and ordered outcomes

$$\begin{aligned}
 z_k &= kt_k - \sum_{i=1}^T d_{ki}, \quad \text{for } k = 1, \dots, T \\
 z_k - asp_k &\geq \delta, \quad \text{for } k = 1, \dots, T \\
 t_k - \sum_{j=1}^n x_j r_{ij} &\leq d_{ki}, \quad \text{for } k, i = 1, \dots, T \\
 \sum_{j=1}^n x_j &= 1 \\
 x_j &\geq 0, \quad \text{for } j = 1, \dots, n \\
 d_{ki} &\geq 0, \quad \text{for } k, i = 1, \dots, T
 \end{aligned}$$

The nature of the decision variables is discussed below:

- t_k = the k -th worst outcome of the portfolio return, for $k = 1, \dots, T$ (free variable);
- z_k = the sum of the worst k outcomes of the portfolio return, for $k = 1, \dots, T$ (free variable);
- $\delta = \min_{k=1, \dots, T} (z_k - asp_k)$ = the worst partial achievement (free variable);
- $(d_{ki})_{k,i=1 \dots T}$ are non-negative variables. $d_{ki} = 0$ if the portfolio return in scenario i is greater than or equal to the k -th worst outcome t_k ; otherwise, d_{ki} is the difference between the k -th worst outcome t_k and the portfolio return in scenario i ;
- $(x_j)_{j=1, \dots, n}$ are the portfolio weights to be determined (non-negative variables).

The modelling principles introduced above draw upon the results of the Reference Point Method stated in section 3.2. Formally, let $(x^*, t^*, d^*, z^*, \delta^*)$ be an optimal solution of model M1 for some aspiration levels $asp_1, asp_2, \dots, asp_T$, where $x^* = (x_1^*, \dots, x_n^*)$, $t^* = (t_1^*, \dots, t_T^*)$, $d^* = (d_{ki}^*)_{k,i=1, \dots, T}$, $z^* = (z_1^*, \dots, z_T^*)$. Then x^* is a SSD efficient portfolio. In addition, its return distribution is close to the reference distribution, in the following sense:

Case 1: If the reference distribution is dominated with respect to SSD, the return distribution of x^* is in some sense uniformly better - it results from the reference distribution by increasing some cumulative outcomes until SSD efficiency is obtained. Thus, the model improves on the reference distribution. In this case (only), the optimal value of the objective function in M1 is positive.

Case 2: If the reference distribution is non-dominated with respect to SSD, the return distribution of x^* is exactly the reference distribution. In this case (only), the optimal value of the objective function in M1 is zero.

Case 3: If the reference distribution is not attainable (some of its outcomes are too high so that there are no feasible portfolio weights that could yield such a distribution), the return distribution of x^* comes uniformly close to the reference distribution. In this case, the optimal value of the objective function is strictly negative.

Note that the sense of “coming close” to a given reference point is not in the sense of distance minimization, but in the sense of optimization of the achievement function.

Remark 3.31. Model M1 may be regarded as a test for stochastic dominance. Depending on the value of the objective function, we can tell if the reference distribution is SSD efficient or not (among the set of return distributions in the portfolio optimization problem). Thus, in order to check if a certain distribution is SSD efficient or not, we can introduce its cumulative outcomes as reference points in model M1. For example, an optimal value of zero in solving the model implies that the distribution is SSD efficient (in this case, the model just finds the portfolio weights that result in this particular return distribution).

Remark 3.32. Typical examples of how to construct a reference distribution are numerous. Choosing an index’s distribution is the most common. Some authors (see [2]) considered the construction of an index composed of a certain number of assets from the initial basket having the highest expected returns. Alternatively, one can take an existing portfolio acceptable to fund managers, see what outcomes in the distribution (e.g. in the tail) can be improved and create an improved distribution that could be used as a reference distribution. This case is illustrated as a numerical example in section 4, considering a mean-variance efficient portfolio as the initial portfolio.

The fact that there is no “universal recipe” for choosing the aspiration levels should be regarded as an advantage of this method. We can analyze the distribution of the portfolio that we obtain, and, if not satisfactory, we can modify the aspiration levels and obtain a further candidate solution portfolio.

For the model M2, we consider both aspiration (asp_k) and reservation (res_k) levels for (some of) the cumulative outcomes z_k . The partial achievement functions are defined by (5).

In addition to the variables used in model M1, there are another T free variables: δ_k , for $k = 1 \dots T$, representing the values of the partial achievement functions.

$$\begin{aligned}
 \max \quad & \delta + \varepsilon \sum_{k=1}^T \delta_k & (M2) \\
 \text{Subject to:} \quad & \delta_k \geq \delta, \quad \text{for } k = 1 \dots T \\
 & \delta_k \leq \frac{\alpha(z_k - res_k)}{asp_k - res_k}, \quad \text{for } k = 1, \dots, T \\
 & \delta_k \leq \frac{z_k - res_k}{asp_k - res_k}, \quad \text{for } k = 1, \dots, T
 \end{aligned}$$

$$\begin{aligned}
\delta_k &\leq \frac{\beta(z_k - asp_k)}{asp_k - res_k} + 1, \quad \text{for } k = 1, \dots, T \\
z_k &= kt_k - \sum_{i=1}^T d_{ki}, \quad \text{for } k = 1, \dots, T \\
t_k - \sum_{j=1}^n x_j r_{ij} &\leq d_{ki}, \quad \text{for } k, i = 1, \dots, T \\
\sum_{j=1}^n x_j &= 1 \\
x_j &\geq 0, \quad \text{for } j = 1 \dots n \\
d_{ki} &\geq 0, \quad \text{for } k, i = 1, \dots, T
\end{aligned}$$

Model M2 also provides a SSD efficient portfolio. In solving model M2, there are five possible cases, depending on the aspiration and reservation distributions chosen by the decision maker. As in the case of model M1, the “uniform improvement” on the reference point and “uniform closeness” to the reference point is not in the sense of distance minimization, but in the sense of optimization of the achievement function. Uniform improvement means increasing some of the cumulative outcomes of the reference, as required by optimizing the achievement function, until efficiency is obtained. Uniform closeness means decreasing some of the cumulative outcomes of the reference, as required by optimizing the achievement function, until feasibility (and efficiency) is obtained.

Case 1: If the reservation distribution (and hence the aspiration distribution) is not attainable, the optimal value of the objective function is strictly negative. In this case, the model provides a SSD efficient solution whose return distribution is uniformly close to the reservation distribution.

Case 2: If the reservation distribution is SSD efficient (thus the aspiration distribution is not attainable), the optimal value of the objective function is zero. In this case, the model provides a SSD efficient portfolio whose return distribution is the reservation distribution.

Case 3: If the reservation distribution is attainable but not SSD efficient and the aspiration distribution is not attainable, the optimal value of the objective function is between zero and one. In this case, the model provides a SSD efficient solution, whose return distribution improves on the reservation distribution and is uniformly close to the aspiration levels.

Case 4: If the aspiration distribution is SSD efficient (which implies that the reservation distribution is not), the optimal value of the objective function is equal to one. The model provides a SSD efficient portfolio whose return distribution is the aspiration distribution.

Case 5: If the aspiration distribution (and hence the reservation distribution) is not SSD efficient, the optimal value of the objective function is greater than one. In this case, the model provides a solution, whose return distribution improves on the aspiration distribution.

Remark 3.33. The reference levels, if achievable, may be seen as a stochastic dominance constraint, as introduced in [2]. Both models (the model presented here and that presented in [2]) have, as a starting point, the assumption that a reference distribution is available. However, the conceptual structures and the scopes of the models are different. The model presented in [2] produces a portfolio whose return distribution dominates the reference distribution with respect to SSD and has a maximum expected return under this condition. Our model produces a portfolio, which dominates not only the reference, but is non-dominated with respect to SSD. Furthermore, if the reference levels are not achievable, our model does not result in infeasibility, but produces a solution that comes close to the reference. In other words, it does as well as it can. This is due to the generalized goal programming approach used in our model: the second constraint of M1 does not require that all cumulated outcomes be greater than or equal to their aspiration levels (δ can be negative, and it is, if the reference is not achievable), but just expresses the worst partial achievement.

3.4. Connection with mean-risk models

3.4.1. Mean-CVaR models. The k -th objective function z_k in (2) represents the sum (or, equivalently, the mean) of the worst k outcomes, for $k = 1, \dots, T$. By attaching aspiration points only for the k -th (with $k < T$) and the T -th objective function, we obtain a mean-CVaR model, with CVaR at confidence level k/T .

The classical mean-CVaR model, with CVaR at confidence level k/T , minimizes CVaR using the optimization formula of Rockafellar and Uryasev ([20]), subject to conditions on expected return. Since CVaR at confidence level k/T is minus the mean of the worst k outcomes, the classical mean-CVaR model is equivalent to:

$$\begin{aligned}
 \max \quad & z_k = kt_k - \sum_{i=1}^T d_{ki} \\
 \text{Subject to:} \quad & z_T = \sum_{i=1}^T \sum_{j=1}^n r_{ij} x_j \geq d \\
 & t_k - \sum_{j=1}^n r_{ij} x_j \leq d_{ki}, \text{ for } i = 1, \dots, T \\
 & \sum_{j=1}^n x_j = 1 \\
 & x_j \geq 0, \text{ for } j = 1, \dots, n \\
 & d_{ki} \geq 0, \text{ for } i = 1, \dots, T
 \end{aligned}$$

where d is the desired level for the sum of all outcomes (or, equivalently, where d/T is the desired level of expected return).

Basically, we have a multi-objective problem with two objective functions to maximize: z_k and z_T , and two different methods for producing an efficient solution.

The reference point approach introduces aspiration levels for both z_k and z_T and then maximizes the achievement function:

$$\max \quad (\min\{z_k - asp_k, z_T - asp_T\} + \varepsilon(z_k - asp_k + z_T - asp_T)),$$

whereas, in classical mean-CVaR optimization, z_k is maximized whilst a lower limit is imposed on z_T : $z_T \geq d$.

Consider now the case where $asp_T = d$. Whether or not both approaches will produce the same solution depends on the aspiration level asp_k . Both solutions will be efficient, and, if asp_k is sufficiently low, they will be the same, with $z_T = d$ and the best possible value for z_k . If asp_k is too high, a different solution will be obtained with the reference point method, in which the value of z_k will be improved at the expense of decreasing the level for z_T . This solution will be efficient in a mean-CVaR model with a lower level imposed for the expected return; it has a better CVaR and a worse mean. Computationally, both models are of similar complexity.

The advantage of the reference point method is that it can deal with more than two reference levels. Although computationally more demanding, the distributions of portfolio returns can be modelled in a more refined way. In addition, this approach can produce any SSD efficient solution in the sense that there exist aspiration (and reservation) levels such that the solution maximizes the corresponding achievement function.

It may be noted that if the k -th objective function in (2) is maximized, a particular efficient solution of (2) is obtained: a portfolio that minimizes CVaR at confidence level k/T . In a mean-CVaR model, with CVaR at confidence level k/T , this corresponds to the lower extremity of the efficient frontier - the minimum risk portfolio.

3.4.2. The maximum mean portfolio. The case of maximizing the last objective function in (2) corresponds to maximizing the expected return in a mean-risk model, irrespective of the risk measure used. In this case, the solution corresponds to the upper extremity of the efficient frontier in any mean-risk model.

3.4.3. Weighted CVaR. It is well known that, for a multi-objective linear problem, we can obtain an efficient solution by the "weighted-sum approach" (see [14], [24]).

Consider the multi-objective linear problem:

$$\max \quad \{Cx \mid x \in Q\},$$

where C is the $T \times n$ criterion matrix, Q is the set of feasible solutions.

Let Λ denote the set of weighting vectors:

$$\Lambda = \{\lambda \in \mathbf{R}^T \mid \lambda_i > 0, \sum_{i=1}^T \lambda_i = 1\}$$

An optimal solution of the single objective linear problem:

$\max \{\lambda^{*T} Cx \mid x \in Q\}$, with $\lambda^* \in \Lambda$ is an efficient solution of the initial multi-objective problem.

The converse is also true: if $x^* \in Q$ is an efficient solution of the initial multi-objective problem, then there exists $\lambda^* \in \Lambda$ such that x^* is an optimal solution of the single objective problem $\max \{\lambda^{*T} Cx \mid x \in Q\}$ (see [24]).

The objective functions in (2) can be considered as minus CVaR at confidence levels $1/T, 2/T, \dots, k/T, \dots, T/T$. Thus, every efficient solution of (2) can be considered as one that minimizes a “weighted CVaR” measure: $\lambda_1 CVaR_{1/T} + \lambda_2 CVaR_{2/T} + \dots + \lambda_T CVaR_{T/T}$.

Remark 3.41. A weighting approach for solving the portfolio optimization problem via multi-objective problem (2) was suggested in [14], [18]. The advantage of using the reference point method instead of this is that the reference points provide a better way of controlling the selection of a solution which is meaningful for the decision-maker, whereas the values of the weights to be attached are more difficult to interpret.

3.4.4. Modelling the tail of distribution. It is possible to set reference levels only for the tail of the portfolio return distribution (i.e. the worst outcome, the sum of the worst two outcomes, ...). The specified reference values are more likely to be achieved, and, in addition, the process is computationally simpler.

Alternatively, we can impose (hard) constraints on VaR at a specified confidence level $\alpha \in (0, 1)$. Suppose α lies between $\frac{k-1}{T}$ and $\frac{k}{T}$. In this case, the value of the variable t_k in model M1 will be minus VaR at confidence level α . Thus, an additional constraint such as $t_k \geq d$ may be added, where d is a desired lower limit for the value of the k -th worst outcome (or VaR at confidence level α). Of course, if the value of d is too high, infeasibility will result.

If a high reference point is used for the last objective function in (2) (representing the expected return of the portfolio), an approximation to the corresponding solution of a mean-VaR model will be obtained. This solution probably will not have the best expected return for the specified VaR level, but will be SSD efficient and with other outcomes close to their reference points.

4. Computational results

In this section, we present an experimental analysis of the models M1 and M2 presented in section 3.3. The returns of the portfolios produced by the models are evaluated in terms of their statistics (mean, median, standard deviation, skewness, kurtosis, range, minimum and maximum). Both in-sample and out-of-sample testing are performed. A data set, drawn from the Hang Seng index, was used for testing the models. Logarithmic returns of the 20 stocks that belonged to the index throughout the period 4 January 1993 - 1 December 2003 were considered. The data set consists of monthly returns and has 131 time periods, considered as equally probable scenarios ($n=20, T=131$). For the out-of sample analysis, the behavior of the portfolios obtained was examined over the eight months following the date of selection (January 2004–August 2004). The models were solved using CPLEX 9.0 optimization solver and the value of ε in the models was fixed at 0.00005.

In solving model M1, there are three cases, depending on the aspiration distribution (see section 3.3). We examine each of these situations, by considering appropriate aspiration distributions. The return distribution of the portfolio obtained in each case is analyzed. In cases where the distribution obtained was considered unsatisfactory, we

Table 1. Statistics for the return distributions of P1 and Hang Seng index (case 1)

	portfolio	
	P1	index
Mean	0.00727	0.00590
Median	0.01168	0.00941
St. Deviation	0.06347	0.08612
Skewness	0.02398	-0.08639
Kurtosis Excess	0.99008	2.15394
Range	0.38901	0.61276
Minimum	-0.16884	-0.34824
Maximum	0.22018	0.26452

introduced reservation levels for some of the outcomes (mainly, in the left tail) and ran model M2. We analyzed the differences in the return distributions obtained with each model.

Case 1: Here we examine the situation where the reference distribution is not SSD efficient. We chose the return distribution of the Hang Seng index as the reference distribution: the 131 monthly returns of the index were ordered and after that cumulated, in order to obtain the sum of the worst k outcomes, for $k = 1, \dots, 131$. The cumulative outcomes of the index were introduced as aspiration levels in model M1. On solving the model, we obtain a positive optimal value of the objective function (0.185), which indicates that the index is not SSD efficient, in contrast with the portfolio obtained by the model. We denote this portfolio as P1.

Table 1 shows the in-sample performance of the two portfolios considered, with P1 outperforming the index on seven of the eight measures used: higher mean, higher median, lower variance, higher skewness (P1 has positive skewness, while the index has negative skewness), lower kurtosis, lower range, higher minimum but with lower maximum.

In Figure 3 we represented the cumulative distribution functions for returns of P1 and the Hang Seng index. In principle, these are "step functions", but in Figure 3 they are displayed as piece-wise linear functions by connecting the chosen bins of the outcomes and their corresponding probabilities. It is obvious that P1 has a cumulative distribution function that is "smaller" for low outcomes. Thus, at low levels, its outcomes are distributed further to the right than those of the index. This was to be expected, since P1 dominates the index with respect to SSD.

For an out-of-sample evaluation, we considered the next eight time periods following the date of selection (January 2004–August 2004). The results are illustrated in Figure 4 and the ex-post parameters are presented in Table 2. The differences in the ex-post parameters for these two portfolios are very similar to those of the in-sample analysis: P1 has almost all the ex-post parameters better than those of the index.

Case 2: Here we examine the case where the reference distribution is SSD efficient. The reference distribution is chosen to be the return distribution of the asset with the highest expected return. This portfolio, composed of just one asset, is efficient in any mean-risk model, irrespective of the risk measure - it lies at the upper extreme of the efficient frontier. Most importantly, it is also SSD efficient - it maximizes the last objective function in (2), representing the sum of all outcomes. We call this portfolio Mean_max.

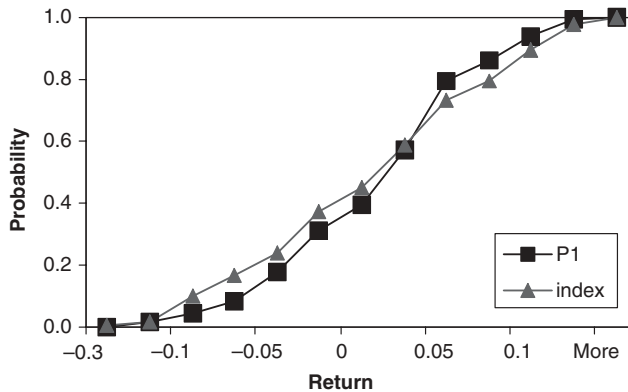


Fig. 3. Cumulative distribution functions for returns of P1 and Hang Seng index

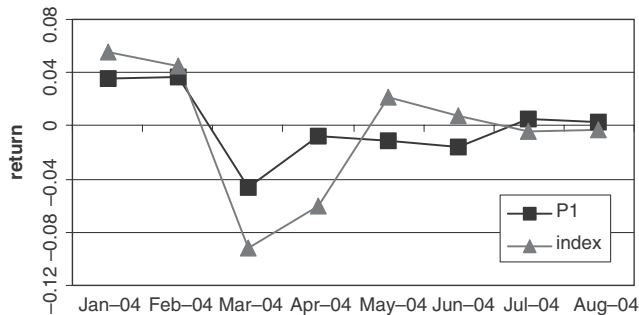


Fig. 4. Out-of-sample evaluation of the Hang-Seng index and portfolio P1 (Jan–Aug 2004)

Table 2. Parameters for the out-of-sample performance of P1 and Hang Seng index (Jan–Aug 2004)

	Portfolio	
	P1	index
Mean	−0.00002	−0.00378
St. Deviation	0.02727	0.05018
Skewness	−0.06427	−0.79758
Kurtosis Excess	0.02027	−0.05637
Minimum	−0.04586	−0.09224
Maximum	0.03631	0.05518
Range	0.08217	0.14742

Running model (M1) having as aspiration points the cumulated outcomes of this distribution, we obtain, as expected, an optimal objective value of 0 and the same portfolio weights: a value of one corresponding to the asset with the highest expected return and zero for the remaining weights. This distribution is SSD efficient and therefore cannot be improved. In order to illustrate this case we performed an "inverse" procedure by specifying the outcomes for the distribution and then determining the portfolio weights that produce these outcomes.

Table 3. Statistics for the return distributions of Mean_ max and P2 (case 2)

	Portfolio	
	Mean_max	P2
Mean	0.01273	0.00491
Median	0.01604	0.00568
St. Deviation	0.14332	0.06084
Skewness	-1.40261	0.12335
Kurtosis Excess	5.74807	0.81840
Range	0.99809	0.34853
Minimum	-0.72631	-0.14352
Maximum	0.27179	0.20501

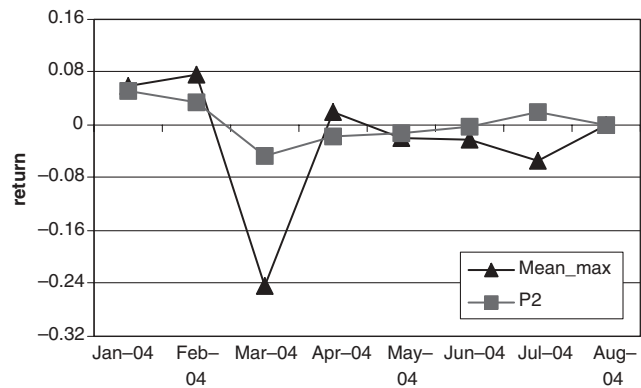


Fig. 5. Out-of-sample evaluation of portfolios Mean_ max and P2 (Jan–Aug 2004)

This distribution is not particularly desirable but was chosen to demonstrate model behavior. Though SSD efficient and also efficient in all mean-risk models, it has undesirable properties, such as a long left tail (worst outcome = -0.72 , second worst outcome = -0.54 , etc.). In such a situation, model M2 can play a useful role. As an example, consider reservation levels for the worst five outcomes to be: $-0.3, -0.3, -0.2, -0.2, -0.2$. Cumulating these values, reservation levels for the first five cumulated outcomes are obtained. Aspiration levels were defined by considering the optimal values for the first five cumulated outcomes and the cumulated outcomes of Mean_ max for the remaining ones. On solving this instance of model (M2), portfolio P2 is obtained and an optimal value of 0.91 , which indicates, as expected, that the reservation distribution is not SSD efficient and the aspiration distribution is not attainable.

As shown in Table 3, although the return distribution of Mean_ max has a higher expected value than that of P2, it has also higher volatility and higher risk of lower returns (negative skewness, high kurtosis, very low minimum).

For an out-of-sample evaluation, the performance of the two portfolios over the eight time periods (January 2004–August 2004) is illustrated in Figure 5 and the ex-post parameters are presented in Table 4. Except for the maximum of out-of-sample returns, which is a little higher for Mean_ max, all the other ex-post parameters are better for portfolio P2.

Table 4. Parameters for the out-of-sample performance of Mean_max and P2 (Jan-Aug 2004)

	Portfolio	
	Mean_max	P2
Mean	-0.02337	0.00303
St. Deviation	0.09914	0.03122
Skewness	-1.81243	0.05663
Kurtosis Excess	4.12961	-0.19501
Minimum	-0.24471	-0.04709
Maximum	0.07535	0.05131
Range	0.32006	0.09841

Table 5. Statistics for the return distributions of P3.1 and P3.2 (case 3)

	Portfolio	
	P3.1	P3.2
Mean	0.00856	0.00324
Median	0.00828	0.00413
St. Dev.	0.06785	0.06142
Skewness	-0.15626	0.23144
Kurtosis Excess	1.02569	0.59529
Range	0.42907	0.32448
Minimum	-0.21090	-0.13603
Maximum	0.21816	0.18845

Case 3: Here we consider the case when the reference distribution is unattainable. We chose as aspiration points the optimal values for the cumulated outcomes. As expected, in solving model M1 with these aspiration points, a negative optimal value for the objective function is obtained (-0.55). The portfolio obtained is denoted by P3.1.

Model M2 can be used to obtain a portfolio whose return distribution has a shorter tail. By defining reservation points for the first seven cumulated outcomes and keeping the same aspiration levels as before (the best values for the cumulated outcomes) and solving model (M2), a portfolio P3.2 is obtained with an optimal value of the objective function between 0 and 1 (0.804), which indicates that the reservation distribution is not SSD efficient and the aspiration distribution is not attainable. (In this example, the reservation levels were chosen by first imposing minimum desirable outcomes for the worst seven outcomes of the portfolio distribution: -0.15, -0.13, -0.13, -0.12, -0.11, -0.11, -0.11. By cumulating these, reservation levels for the first seven cumulated outcomes were obtained).

As shown in Table 5, although the return distribution of P3.2 has a lower mean (and a lower maximum), most of its statistics are better, especially those concerning the distribution's tail (higher skewness, lower kurtosis, lower range, higher minimum).

The out-of-sample returns of portfolios P3.1 and P3.2 over the eight time periods (January 2004–August 2004) are illustrated in Figure 6 and the ex-post parameters are presented in Table 6. The better performance of portfolio P3.2 can be clearly seen. All the ex-post parameters (except for variance, which is slightly higher) are better.

We now present a case study illustrating how our model makes a choice as compared to the mean-risk models. The purpose is to select a portfolio whose return distribution has a high expected value, but in the meantime a satisfactory left tail.

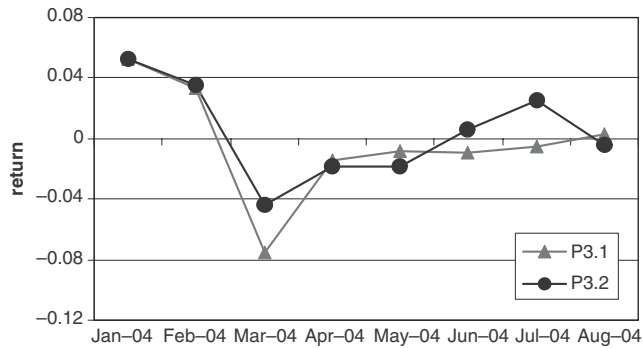


Fig. 6. Out-of-sample evaluation of portfolios P3.1 and P3.2 (Jan–Aug 2004)

Table 6. Parameters for the out-of-sample performance of portfolios P3.1 and P3.2 (Jan–Aug 2004)

	Portfolio	
	P3.1	P3.2
Mean	−0.00307	0.00421
St. Deviation	0.03753	0.03207
Skewness	−0.56962	0.09690
Kurtosis Excess	1.73200	−0.80341
Range	0.12756	0.09674
Minimum	−0.07525	−0.04407
Maximum	0.05231	0.05267

We first consider the mean-variance efficient portfolio with an expected return $d_1 = 0.00872$, which is quite close to the maximum expected return ($d_{max} = 0.01272$). Denote this portfolio by P_{var} . An analysis of the return distribution of P_{var} reveals an unsatisfactory left tail. At this mean level, imposing a CVaR constraint does not lead to a significant improvement in the left tail of the portfolio return. That is because, for high mean levels, the efficient portfolios in mean-risk models are quite similar, irrespective of the risk measure used. CVaR at confidence level 0.95 of P_{var} is 0.1480, while the best 0.95CVaR for this expected value of the portfolio return is 0.1479. Imposing an additional constraint on CVaR would result in a portfolio hardly different than P_{var} . Thus, we shall instead just consider the mean-CVaR efficient portfolio for the mean level $d_1 = 0.00872$. Denote this portfolio by P_{CVaR} . This portfolio is also very similar to P_{var} . Thus, the return distributions, and obviously the left tail, are very similar.

The model M1 is used to shape the return distribution of a "desirable" portfolio P4 in the following way. Firstly, we set the aspiration level for the expected value of the return $d_1 = 0.00872$ (as achieved by P_{var} and P_{CVaR}). Secondly, 20 optimization runs are carried out in respect of the worst 20 cumulative outcomes of the portfolio return. These 20 optimal values are set as aspiration levels for the worst 20 cumulative returns, together with the original mean return aspiration level. The portfolio P4 is computed using model M1 with these 21 aspiration levels.

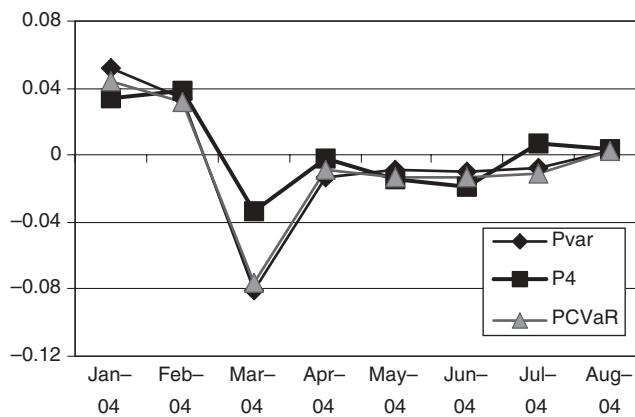
The mean return of P4 is lower than those P_{var} and P_{CVaR} , but only marginally (since we don't necessarily require a level of mean return, but just regard it as a target).

Table 7. The left tails of return distributions of P_{var} , P_{CVaR} and P4 as described by the worst 20 outcomes

	Ordered outcomes									
	1	2	3	4	5	6	7	8	9	10
P_{var}	-0.210	-0.175	-0.152	-0.125	-0.123	-0.123	-0.111	-0.109	-0.101	-0.100
P_{CVaR}	-0.202	-0.175	-0.156	-0.130	-0.126	-0.122	-0.105	-0.105	-0.105	-0.100
P4	-0.174	-0.166	-0.147	-0.117	-0.116	-0.109	-0.093	-0.085	-0.079	-0.072
	Ordered outcomes									
	11	12	13	14	15	16	17	18	19	20
P_{var}	-0.098	-0.092	-0.089	-0.073	-0.072	-0.062	-0.060	-0.056	-0.054	-0.052
P_{CVaR}	-0.099	-0.094	-0.086	-0.085	-0.070	-0.056	-0.055	-0.053	-0.053	-0.053
P4	-0.069	-0.068	-0.067	-0.062	-0.059	-0.058	-0.056	-0.055	-0.052	-0.052

Table 8. Statistics for the return distributions of P_{var} , P_{CVaR} and P4.

	Portfolio		
	P_{var}	P_{CVaR}	P4
Mean	0.008726	0.008726	0.007535
Median	0.008729	0.01108	0.010040
Standard Deviation	0.069247	0.069526	0.062459
Skewness	-0.166730	-0.140564	0.038810
Kurtosis	0.902741	0.827794	1.345610
Range	0.428051	0.417571	0.397412
Minimum	-0.210489	-0.201803	-0.173714
Maximum	0.217563	0.215768	0.223699

**Fig. 7.** Out-of-sample performance of portfolios P_{var} , P_{CVaR} and P4 (Jan–Aug 2004)

This is considered to be an acceptable trade-off since the improvement in the left tail of the P4 return distribution (as compared with P_{var} and P_{CVaR}) is remarkable.

The in-sample parameters for the return distribution of these portfolios are presented in Table 8.

The out-of-sample results, for the eight months following the date of selection, are presented in Figure 7 and in Table 9. P4 has an obviously better out-of-sample performance.

Table 9. Parameters for the out-of-sample performance of portfolios P_{var} , P_{CVaR} and P4 (Jan–Aug 2004)

	Portfolio		
	P_{var}	P_{CVaR}	P4
Mean	−0.004191	−0.005356	0.001816
St. Deviation	0.039118	0.035921	0.024856
Skewness	−0.698098	−0.720870	0.334644
Kurtosis Excess	1.983224	1.821935	−0.761385
Range	0.133357	0.120194	0.071497
Minimum	−0.080971	−0.075778	−0.033089
Maximum	0.052385	0.044416	0.038408

5. Concluding remarks

The portfolio selection problem is presented as a particular case of decision making involving random variables. Our purpose is to make meaningful decisions, according to observed economic behavior, and at the same time preserve computational tractability. Meaningful decisions are usually considered to be those consistent with an axiomatic model of preference (stochastic dominance). Here, we considered the problem of choosing a portfolio that is efficient with respect to second order stochastic dominance (SSD). Thus, decision makers are assumed to be rational and risk-averse. The number of portfolios that are efficient with respect to second order stochastic dominance is infinite and additional criteria are needed in order to select a specific, meaningful solution.

We proposed to do this by specifying a reference distribution of return outcomes that would be desirable under various scenarios, and finding the portfolio whose return distribution comes closest to the reference.

The models use target levels for the sums of the worst k outcomes, for $k \in \{1, \dots, T\}$, where T is the number of (equally probable) scenarios considered. The first model uses only aspiration levels. The second one uses, in addition, reservation levels, which pre-empt the aspiration levels in the optimization. Both models produce SSD efficient portfolios irrespective of the reference distribution used.

The models were evaluated using a dataset drawn from the Hang Seng index and using different reference distributions. Both in sample and out-of-sample analyses were performed. The portfolios produced by the models were evaluated by comparing statistics of their return distributions. Of primary interest in the evaluation is the status of the reference distribution with respect to SSD efficiency. In cases where the reference distribution was not SSD efficient, the portfolio produced by the first model M1 had a return distribution with better statistics than the reference, and, in addition, better out-of-sample results. When the reference distribution was SSD efficient, model M1 produced a portfolio whose return distribution was exactly the reference. The robustness of the model is such that, when the reference distribution is unattainable, a portfolio is produced whose return distribution is as close as possible to the reference.

Model M1 was extended in order to gain greater modelling power in selecting desirable portfolios. In cases where the return distributions of the portfolios produced have undesirable outcomes under specific scenarios, the second model M2 can be used to produce portfolios, which improve on these outcomes. In particular, we used model M2 to model the left tail of the return distributions. The effectiveness of model M2 was

demonstrated by producing portfolios that outperformed those produced by M1, both in terms of in-sample statistics and out-of-sample results.

It is envisaged that, as in many multi-objective applications, the models are used interactively with the decision maker specifying new aspiration (and reservation) levels in order to obtain portfolios with progressively better return distributions.

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